Data Understanding

1. Number of provided datasets

   We have 9 datasets witch are: Customers, Products, Orders, Stores, Returns, Employees, Suppliers, Inventory, Order_Items

2. dataset explanations

   Customers, contains all the information about customers
   Products, contains all the information about Products
   Orders, contains all the information about customers Orders
   Stores, contains all the information about Stores
   Returns, contains information about returned items from an order
   Employees, contains all the information about Employees and the stores where they work
   Suppliers, contains all the information about Suppliers
   Inventory, lets you see the products available in the store
   Order_Items, contains all the information about Order_Items

3. Primary Key of each table

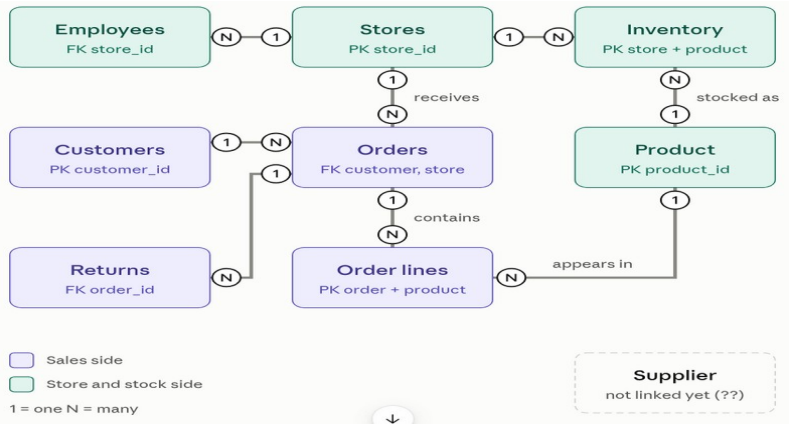
Tables	**Primary keys**
Customers	Customer_ID
Products	Product_ID
Orders	Order_ID
Stores	Store_ID
Returns	Return_ID

Tables	**Primary keys**
Employees	Employee_ID
Suppliers	Supplier_ID
Inventory	Inventory_ID
Order_Items	Order_Items

4. Possible Foreign Key

Tables	**Foreign keys**
Returns	Order_ID
Employees	Store_ID
Orders	Customer_ID, Product_ID, Store_ID
Inventory	Product_ID, Store_ID
Order_Items	Order_ID, Product_ID

5. Relationship between the tables



Excel Formulas

1. Total Revenue : 2496824383
2. Total Quantity Sold : 49432
3. Average Selling Price : 36244,3068
4. Average Order Value : 151057,1954
5. Highest Sales Amount : 499915
6. Lowest Sales Amount : 1014
7. Number of Orders : There are 50000 Orders
8. Number of Customers : There are 5000 Customers
9. Number of Products : There are 500 Products
10. Number of Stores : There are 25 Stores

Business Questions

1. Which product generated the highest revenue?

   Groceries Item 444 generated the highest revenue.

2. Which store performed best?

   The ShopSmart Lagos store achieved the best results.

3. Which category sells the most?

   The Beauty category had the highest sales.

4. Which month generated the highest revenue?

   December was the month with the highest sales revenue.

5. Which customers spend the most?

   Melissa Shaffer

6. Which products are frequently returned?

   We have : Beauty Item 114 and Home Item 263

7. Which supplier supplies the highest number of products?

   Supplier is not related to any other table.

8. Which stores have low inventory?

   ShopSmart Kaduna

9. Which region performs best?

   The southern region performed the best.

10. Which product category should management invest in?

    Beauty category

Business Recommendations

Three key findings

- **Beauty is the leading category, and Lagos and the south lead geographically.** Beauty has the highest sales, ShopSmart Lagos is the best store, and the southern region performs best. Revenue is concentrated in a few places and a few categories.
- **Sales are seasonal.** December is the highest-revenue month, so a large share of the year's revenue depends on one peak period.
- **Customers are active and repeat buyers.** 5,000 customers placed 50,000 orders, about 10 orders each. That is roughly 499,000 revenue per customer (2,496,824,383 ÷ 5,000), so retention matters more than acquisition.

Three business recommendations

- **Invest in the Beauty category.** Increase stock depth and range, and negotiate better supplier terms for it. Before doing so, fix its return problem (see the risk below).
- **Rebalance inventory by store.** ShopSmart Kaduna has low stock. Move stock from slower stores to it, and set reorder points for each store so best sellers like Groceries Item 444 don't run out.
- **Plan for December.** Build stock, staffing and promotions around the peak. Use quieter months for loyalty offers aimed at your top spenders, such as Melissa Shaffer.

One risk identified

Beauty Item 114 is frequently returned, and Beauty is also the category you want to invest in. High returns reduce real revenue and add handling costs. Investing heavily in Beauty without first finding out why it is returned (quality, wrong description, damage) could erode the profit you expect from it. Home Item 263 shows the same pattern.

One opportunity identified

Close the gap between the south and the other regions. The south performs best and ShopSmart Lagos leads. Copying its practices (product mix, pricing, store layout, promotions) in underperforming stores and regions, such as the north where Kaduna is, could raise total revenue without opening any new stores